IRAS Research Data Management Guidelines

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Preface

This book serves as a guideline for research data management at the Institute for Risk Assessment Sciences (IRAS) of the Faculty of Veterinary Science. It outlines and defines a set of principles regarding research data and provides practical handlers for researchers in accordance with the regulations established in the University Policy Framework for Research Data from Utrecht University (UU) and the Research Data Management Guidelines developed by the Faculty of Veterinary Medicine. These guidelines can be used as a reference together with existing UU policies, faculty guidelines and/or procedures already established by research groups.

> **!** Important
>
> Please note that the UU's University Policy Framework for Research Data dates back to 1 January 2016. This is before e.g. the GDPR was established. Hence, some of the information mentioned in the policy framework is not up to date. UU is in the process of creating a new policy framework. In the meantime, please consult RDM Support in case of doubt about the information in the current policy framework.

Part I

Introduction

The current chapter describes:

- Chapter 1: how this book is structured.
- Chapter 2: why good research data management is important.
- Chapter 3: responsibilities of everyone involved in the research project.
- Chapter 4: where you can find support.

1 Structure of this book

The IRAS Research Data Management guidelines are structured according to the Research Data Life Cycle (Figure 1.1). This cycle can be broadly divided into three phases: before (i.e. planning and designing), during and after research. Research data management (RDM) activities and actions take place during all three phases.

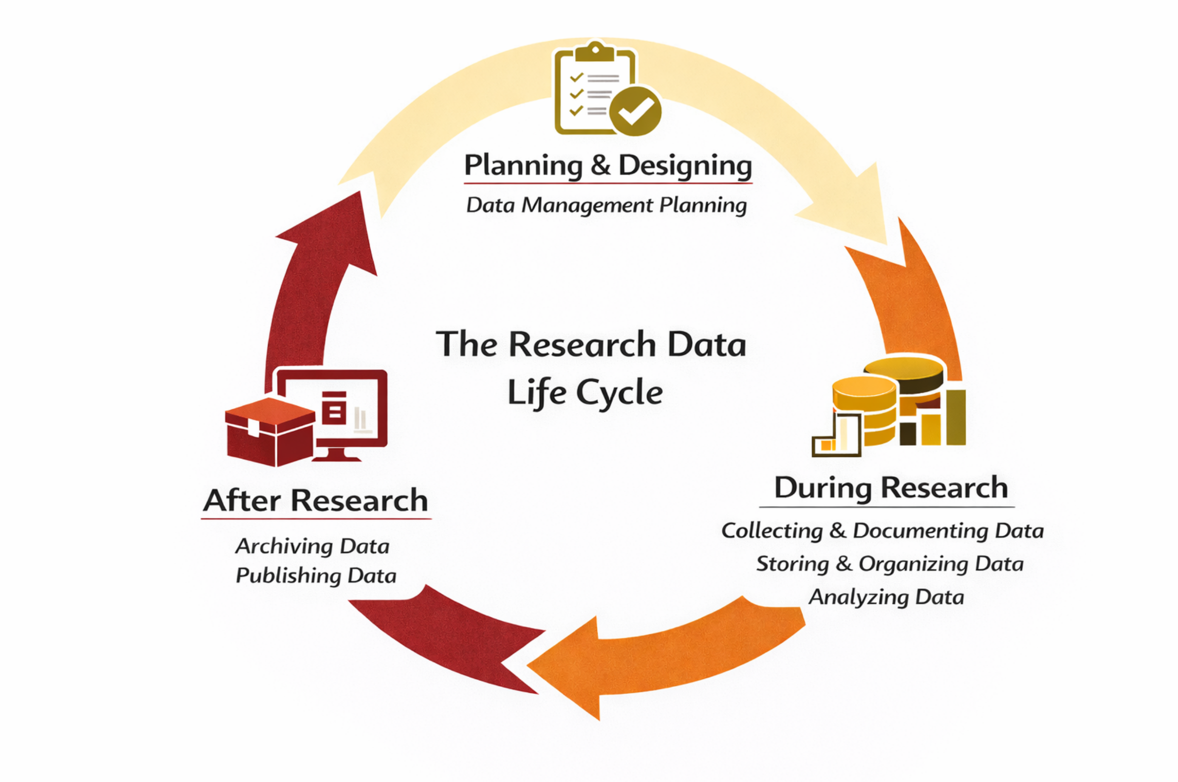

Figure 1.1: The Research Data Life Cycle

2 Importance of good research data management (RDM)

RDM refers to how you handle, organize, and structure your research data throughout the research process. Good RDM is essential for ensuring the integrity and validity of research, facilitating reproducibility and promoting collaboration. It also helps protect sensitive information and comply with regulations. Key reasons why good RDM is important include:

- **Ensuring data quality and security:** proper RDM prevents data loss, protects against unauthorized access, and ensures that research results are reliable and valid.
- **Increasing efficiency:** organizing, documenting, and structuring data saves time and resources by making it easier to find, understand and use data. Not only for others, but also for your future self.
- **Enabling reproducibility and validation:** well-managed, documented data allows other researchers to verify findings, contributing to the overall reliability and validity of scientific knowledge.
- **Allowing for reuse and collaboration:** making data understandable for others facilitates collaboration by enabling researchers to easily share and access data, accelerating scientific discovery.
- **Compliance and integrity:** good RDM ensures compliance with legal (e.g. GDPR when working with personal data), ethical, and funder requirements, and adheres to scientific integrity standards.
- **Improving visibility and impact:** making data FAIR (Findable, Accessible, Interoperable, Reusable) increases citation potential and the overall impact of research.

Proper RDM covers the entire data life cycle, from initial planning and collection to storage, analysis, archiving, and sharing. It is essential for effective Open Science and FAIR practices.

2.1 Open Science

In 2017, in order to accelerate and improve the realization of research results and their societal impact, the UU decided to make the transition to Open Science. Open Science is about increasing the reuse of research, and making sure that publicly funded research is accessible to all. Key to achieving this is adhering to the FAIR principles: ensuring the findings and data behind research results are findable, accessible, interoperable, and reusable (Bruce and Cordewener (2018)).

2.2 FAIR principles

In line with the UU's commitment to Open Science, research data at IRAS is expected to follow the FAIR principles:

- **F - Findable:** data and metadata should be findable, preferably digitally.
- **A - Accessible:** data and metadata should be stored and maintained in such a manner that they are accessible and with clear terms of use.
- **I - Interoperable:** data and metadata data should be stored and maintained in such a manner that the data can be easily opened, exchanged and combined with other datasets.
- **R - Reusable:** data are licensed and described in a manner that facilitates its reuse by other users.

Guidance on how to make your research data FAIR is highlighted in the recommendations provided below. To learn more see the RDM Support guide on how to make your data FAIR.

3 Responsibilities

The UU distinguishes between different individuals or (groups of) actors, each bearing specific responsibilities for research data management. Specific responsibilities are outlined in the UU's University Policy Framework for Research Data. Here below, we've outlined our own additional specifications for research data management at IRAS.

> ❗ Important
>
> Please note that the UU's University Policy Framework for Research Data dates back to 1 January 2016. This is before e.g. the GDPR was established. Hence, some of the information mentioned in the policy framework is not up to date. UU is in the process of creating a new policy framework. In the meantime, please consult RDM Support in case of doubt about the information in the current policy framework.

3.1 Project leader

The **project leader** (e.g. principal investigator) is the main person responsible for:

- Ensuring that all project members (see below) abide by the UU's University Policy Framework for Research Data, the Research Data Management Guidelines of the Faculty of Veterinary Medicine and the IRAS Research Data Management Guidelines (i.e. this document).
- Organizing proper handling of research data.
- Making sure that research data is stored and preserved in a sustainable manner after the research project has ended.
- Keeping track of what type of identifying personal data will be collected and ensuring that the confidentiality is guaranteed through a detailed description of how identifying personal data will be handled, e.g. in a privacy statement or privacy regulations document. An example of a privacy statement can be found here: https://obokids.nl/privacy-2/. The description can be kept fairly simple and general if the identifying personal data is kept in the central directory, but may need to be more elaborate and specific when an alternative is wanted or needed.

3.2 Project members

Depending on their role and the project requirements, **project members** (e.g. students, researchers, research leaders, research assistants, data managers, project managers, etc.) have certain responsibilities as well. The creators of the data (researchers, students and their supervisors) have primary responsibility for good stewardship of their research data. For students, it is the responsibility of the thesis supervisor to ensure that research data is stored and preserved in a sustainable manner after the project has ended.

More specifically, project members should:

- Familiarize themselves with the UU's University Policy Framework for Research Data, the Research Data Management Guidelines of the Faculty of Veterinary Medicine and the IRAS Research Data Management Guidelines (i.e. this document).
- Sign the IRAS Identifying Personal Data Confidentiality Agreement in case the project requires working with personal data (see Appendix X).
- Draw up proper procedures and design processes for the collection, storage, use, reuse, availability and long-term archiving of and provision of access to research data within the research project. The best way to do this is by developing a Data Management Plan (see Chapter 5) based on the information available in these guidelines.
- Factor in the costs for the preservation and management of the research data and how they will be covered (institute-, subsidiary-funding or otherwise) (see Chapter 6).
- In the case of external data, agreements must be made with the provider of the external data. If participating in a joint study or research project agreements on data management should be made and recorded.
- Ensure that the integrity, confidentiality and protection of the data are guaranteed (see Data security and privacy).
- Make sure that research using personal data (which is the case for most research projects at IRAS) is conducted according to the General Data Protection Regulation (GDPR, in Dutch: AVG). Other laws, such as the Wet Medisch wetenschappelijk Onderzoek (WMO) and Wet Geneeskundige Behandelingsovereenkomst (WGBO), apply alongside it, filling in specific gaps or adding stricter rules in certain areas. In addition, there are specialized codes of conduct for handling personal data in research, for example the COREON Code of Conduct for Health Research. For more information, see the RDM Support page on policies, codes of conduct and laws and Data security and privacy.
- Make arrangements for the continued operational management of the research data after the end of the research project or termination of the employment contract at the institution.

An extended version on research data management responsibilities can be found in the UU's University Policy Framework for Research Data.

4 Support

4.1 IRAS Data Management team

The IRAS Data Management team is available for consultation on matters related to data management. We have a wide variety of expertise within the team, from data collection and cleaning to working with GIS data and setting up and hosting applications. While capacity within the team is limited when it comes to providing hands-on support, we will most of the time be able to provide advice on any data related questions you might have or bring you in to contact with the right people to help you further.

> **IRAS Data Management team**
>
> Data management representatives:
>
> - **Coordinator**: Zimbo Boudewijns
> - **One Health Chemical (OHC) representatives**: Inka Pieterson, Marthe Knijff (iras_dm@uu.nl)
> - **One Health Microbial (OHM) representative:** Peter Scherpenisse (p.scherpenisse@uu.nl)
> - **One Health Toxicology (OHT) representative:** Kas Adriaans (k.j.adriaans@uu.nl)
> - **Veterinary and Comparative Pharmacology (VCP) representative:** André Joubert (a.j.joubert@uu.nl)
>
> Other expertise:
>
> - **GIS specialists**: Lloyd Roga, Carmel Suchard
> - **Data solution architect and application specialist**: Adam El Kassimi

4.2 UU RDM Support team

In addition to the data management team at IRAS, the UU's Research Data Management (RDM) Support team is there to support, advise and assist all researchers at UU with data management, data analysis, research software and code. A multidisciplinary network of data experts, software specialists and research engineers offer their expertise through training, tools, infrastructure, consultancy, and custom-made solutions at every stage of a research project. The support they provide complies with the FAIR principles: findable, accessible, interoperable and reusable. That way, it contributes to Open Science.

> **i** RDM Support workshops and guides
>
> To learn more about how to manage your research data and software, the UU's RDM Support team offers several trainings and workshops, as well as walk-in events. For a complete event agenda, together with information on trainings and workshop, please refer to the RDM Support website.
> Reading guides on diverse RDM topics can also be found on the RDM Support website. The same goes for tools that may be useful.

4.3 Other support services

Besides the IRAS Data Management team and RDM Support team, the following support services exist at UU:

- Research Support Office (RSO) of the Faculty of Veterinary Medicine: each faculty has an RSO. The RSO supports researchers throughout the year in the entire process of acquiring grants. For example: advice, information, reading, examples, budget assistance, data management, ethical questions, administrative data, etc.
- Privacy Officers of the Faculty of Veterinary Medicine: if you are starting a new project or research involving the use of personal data, exchanging data with a third party, or have questions about the secure storage of data or GDPR, please contact the Privacy Officers. You could also contact them for questions about the safe and responsible use of AI.

4.4 Contact

In case you have questions about research data management, please reach out!

> **i** Contact details
>
> IRAS Data Management team: see Section 4.1
> RDM Support team: info.rdm@uu.nl
> RSO of the Faculty of Veterinary Medicine: rso-dgk@uu.nl
> Privacy Officers of the Faculty of Veterinary Medicine: privacy-dgk@uu.nl

Part II

Planning and designing

The current chapter describes:

- Chapter 5: why you should write a DMP and how to do that.
- Chapter 6: why you should estimate the costs that will be associated with data management and how to do that.

5 Data management plan (DMP)

RDM Support has instructions for developing a DMP. At this page several questions about DMPs are answered, such as the ones below.

Questions and answers

- **What is a DMP?** A DMP is a formal document you develop at the start of your research project which outlines all aspects of managing your data, both during and after your project.
- **Why do you need a DMP?** Data management is all about making your work efficient and creating more value for your data for yourself and others, during and after your research, for instance by working towards FAIR (findable, accessible, interoperable, reusable) data.
- **How to write a data management paragraph for a grant proposal?** When writing a research grant proposal, funders, such as NWO, will often ask for a data management paragraph.
- **How to write a DMP?** When writing your own DMP, you can:
  - Use DMPonline to write your DMP: DMPonline contains templates from many different funders (e.g., NWO, ZonMw and ERC) as well as the institutional templates from participating universities. The template offered by Utrecht University is tailored to the requirements which are laid down in the University Policy Framework for Research Data And approved by both NWO and ZonMw.
  - Get advice or feedback: if you are in need of advice or want your DMP reviewed, you can contact Research Data Management Support anytime, or click "Request feedback" in DMPonline.
  - Need more information or an introduction? Follow the online training Learn to write your DMP or join the workshop Quick start to Research Data Management and ask all your questions.

The use of DMPonline is further explained at this page.

5.1 Our recommendation

Our recommendation is to write a DMP at the start of your research project and use the applicable template in DMPonline. This contains several examples you can use for your project. You can also find an example of a DMP developed for a research project at IRAS here: **DMP OBO2 WP6 06102025.pdf**.

If you have any questions or want to have your DMP reviewed, please contact RDM Support as outlined above. During onboarding this will be explained to new staff.

6 Costs

Research data management will likely incur financial costs. In particular, the costs of data storage, archiving and publishing can be high. In some cases, these costs may not be covered by the university, and thus they will need to be obtained through funding or via other routes during the planning phase of the project. Luckily, most funders consider the costs for data management eligible for funding.

> Support
>
> RDM Support has developed a **guide to estimate the costs of data management** for each research phase and research activity. The costs of research data management will be a lot lower when you plan well ahead, instead of applying it retrospectively, both in terms of time and money.
> Check out the RSO intranet page for help with the **funding and grant acquisition process**.

6.1 Our recommendation

Our recommendation is to use RMD Support's guide to estimate the costs involved in research data management before you start your research project (e.g. when you start writing your DMP or applying for funding), because this will save you time and money in the end. If you need support or assistance, please contact RDM Support.

Part III

During research

The current chapter describes:

- Chapter 7: what tools the UU offers to collect research data, how to use them and what to pay attention to in terms of data management.
- Chapter 8: how best to store and organize your research data during your research project.
- Chapter 9: why documenting your data is important and how to do this.
- Chapter 10: where best to analyze your data and how to use GitHub for version control.

7 Collecting data

7.1 Formdesk

...

7.2 Informed consent

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7.3 Qualtrics

If you want to create a survey that will collect personal data, the UU has one option and that is Qualtrics. Qualtrics is free of charge and can be used to set up surveys, collect data, distribute and analyze responses. Issues such as privacy and security are well taken care of. For further information please refer to the Qualtrics intranet page of the UU. It tells more about privacy, rules of conduct, non-standard UU modules in Qualtrics and more.

You can log into the Qualtrics website with your Solis-id (or UU email address), Solis-password and two-factor authentication. Once you have logged into Qualtrics, you can create a new survey via the Projects page. The Project page shows all surveys you have created (or that have been shared with you). You can collaborate on surveys with others by sharing them.

Qualtrics is known for having its fair share of quirks. A lack of awareness of these can lead to frustrations when using it to collect data. Therefore, Appendix B offers recommendations and tips and tricks for building a Qualtrics survey in a FAIR and structured way to enable the collection of high-quality data, while keeping privacy requirements in mind. This may take some investment upfront, but will pay off in the end.

8 Storing and organizing data

8.1 Storing data

Properly storing and organizing active data within a project is very important, and with the growth of the PHS-IRAS division it becomes even more important. Active data is data that will be subject to change throughout your research. Properly structured and annotated data ensures that your data is properly preserved, making it easier to decide what to publish, archive, or discard.

The aim of these guidelines is to make storage more uniform across the different projects and to make sure the tools the UU has to offer for data storage are used optimally. Of course, projects differ in their specific data needs and the goal is not to enforce one specific way of working for all projects. However, by reusing the same principles where possible, findability and reproducibility are greatly improved.

If you want to learn more about data in the final phases of your research, please consult Chapter 11 and Chapter 12.

8.1.1 Why is taking good care of data storage important?

1. Privacy regulations (GDPR) and medical ethical clearance (METC) procedures make it necessary that personal data are carefully handled.
2. With regard to coded (pseudonymized) or anonymous data:
   - It is important that results can be reproduced.
   - It is important that errors will be prevented.
   - If there is a follow-up after the current project, it is necessary that the data have been clearly managed and stored to facilitate reuse.
   - In case of an audit the correct data should be easily obtainable.
3. With regard to both coded (pseudonymized) or anonymous data and identifying personal data:

- When an employee leaves IRAS the data should be accessible, and should be managed and stored in an understandable way for others.
- It is important that no data is lost.
- The storage location must be secured against unauthorized access.

It is important to think about how to organize your data at the start of your research project. Of course, it is possible that during the project you end up having to do some things differently than you had anticipated. However, you should draft plan for storing your data based on the type of data you expect to collect.

This chapter will guide you through the steps to take when making a plan to organize your data. It discusses all the tools that are available at Utrecht University and points you to relevant resources you can use. The advice is to consult the data management representative of your research group/department to discuss the best option for your research project (see Section 4.1).

8.1.2 Tools and information

Within the UU several sources of information are available where to store your data safely. It depends upon the research phase and the type of data what an appropriate storage platform is. This RDM Support Guide tells more about storing data during research. You can check out the UU storage finder to find the correct platform for your data. Below you will also find some concrete recommendations that apply to most research projects within IRAS.

8.1.3 Data classification

To determine the necessary measures to protect your data, it is helpful to first perform a data classification of the data you will collect in your project. This will help you find a suitable data storage tool. The goal of classifying information is roughly categorizing the potential impact for the university if the availability, integrity or confidentiality of the data get compromised.

By answering a set of questions you will determine the value of your data and the security risk that your data is exposed to. You will conclude what the impact is that a breach can have on the availability (what if data cannot be accessed or the data is lost entirely), the integrity (what if the data is incorrect) and confidentiality (what if an unauthorized person gains access to the data or the data is leaked). Making this assessment is called data classification. When you consult the storage finder, it will also ask you about the data classification.

For more information about data classification see the following pages:

- https://www.uu.nl/en/research/research-data-management/guides/legal-instruments-and-agreements

- https://intranet.uu.nl/en/security/information-security-data-classification

8.1.4 Important considerations for storing personal data at IRAS

! Important

1. Personal data should only be accessible to those who need to work with it. This means that data (that is used only infrequently) should be stored centrally (where it can be properly secured) and only accessible to a limited number of people.

2. Personal data should not be stored on a local drive. Apart from unauthorized access, the risk of data loss is much higher with a local drive due to it crashing, getting lost, or being stolen than with a network drive (which is automatically backed up). YODA is also a suitable location to store these data.

3. Personal data may only be accessed on secure UU devices. For example, it is not permitted to use personally identifiable information on personal computers, USB sticks, hard drives, or in the cloud.

It is very important that data (in general) does not end up in the hands of unauthorized persons. Data protection starts at the workplace, where simple measures can greatly reduce the chance that unauthorized persons obtain personal data. Therefore you should always:

1. Keep personal data, like informed consent forms from questionnaires, field forms with names and addresses, etc., under lock and key and separate from the coded (pseudonymized) or anonymous data.
2. It is important that access to PCs on which identifying personal data is processed is adequately protected. Processing of identifying personal data is only allowed on secure UU (portable) computers to which access is secured with a 2FA login-procedure with password, with an automatic log-off after a period of inactivity.

Particular attention deserves employees that work at home. IRAS does only allow accessing or working with identifying personal data at home via VPN. Identifying personal data should not be transported by means of portable mediums. In exceptional cases, for single transport from institute to institute, for instance, together with samples, encrypted information should be carried on a password secured and labeled data stick.

8.1.5 IRAS storage recommendations

The UU offers many types of storage, all of which have different specifications (see also the storage finder). The most commonly used storage options are the O-drive, YODA and OneDrive. While all of these are supported by the UU, it's important to consider the limitations of each of these systems before you decide which one(s) to use for your project. If you collect sensitive data within your project we recommend to use the UU network O-drive to store and analyze your data. YODA is very convenient when you want to collaborate with external partners and it has advanced features for making your data FAIR that are not available on the O-drive. OneDrive can be convenient to use too but it has important downsides as discussed below.

8.1.5.1 OneDrive for business

OneDrive for business is installed on most UU-maintained laptops and is safe to store your data. However, because OneDrive is linked to your UU account the data will be lost when you leave the UU. It is advisable to give more people access to your data (e.g. your supervisor or data managers), even though you can give other people access to your OneDrive folders that other people can no longer access these folders when your UU account no longer exists. Therefore the O-drive and YODA are more suitable for storing research data. OneDrive can be convenient for storing documents because you can collaborate with other people, however it is advisable to store the final versions of these documents on the O-drive or YODA.

> **!** Important
>
> Be very careful with storing data locally on your computer. When the C-drive is linked to the institutional OneDrive, it can be used to temporarily store data. However, this is not always by default the case, which means data stored on the C-drive is not automatically back-upped. In addition, keep in mind that no one else has access to the data stored in your personal OneDrive in case that is necessary, it's therefore advisable to work from a shared folder on the O-drive. Storing data locally without OneDrive is a risk as your data will be lost in case your computer crashes. See the text below for recommendations for storing your data.
> Don't use commercial tools like Dropbox, Google Drive, or the consumer version of OneDrive: your data may be stored outside of Europe (and therefore not covered by European law), and the products are offered "for free" (so the companies monetize your content).

8.1.5.2 O-drive

The O-drive is meant for collaboration with colleagues and research groups within the faculty. Here you can store and share information with your colleagues. It's secure storage provided in a secure folder for which you've set permissions and which you then continue to monitor. Automatic backup of your data is provided in the form of snapshots that go back 90 days. When you plan the data storage for your project, consider if you store (raw) data that you don't use regularly and therefore you might not notice within the 90-day back-up period if they are accidentally deleted or modified. The YODA vault is a suitable place to store raw data that prevents accidental deletion or modification. Finally, the O-drive is not suitable for storing large data files that you don't regularly, as it's very expensive for the faculty. Those data can be best stored in the Yoda vault, see the section on YODA.

On the O-drive it is recommended to create a 'Privacy folder' and a 'Project' folder for your project. Identifying data is stored in the project's 'Privacy' folder, with restricted access only to those who need this data. The pseudonymized research data can be stored in the 'Project' folder, with access for researchers performing the analysis. It is very important that you properly manage access rights to the folders on the O-drive. If necessary, you can create separate folders with restricted access, for e.g. Master's students. Or add a password to a logbook used by multiple people.

> Storing data on the O-drive
>
> Advantages:
>
> - Secure storage
> - Direct access to your data
> - Share data with colleagues within the faculty
> - Separate privacy folder for identifying data
>
> Disadvantages:
>
> - Back-ups are only kept for 90 days
> - No option to give collaborators outside of the faculty access
> - Expensive storage, not preferred for long term storage of large amounts of data
> - Loading large files can be slow when working via the VPN

8.1.5.3 YODA

YODA is suitable for long-term collaboration and long-term storage of research data. Automatic backup is provided and YODA is suitable for a wide variety of data types and formats, including identifying or sensitive personal data.. The Faculty of Veterinary Medicine promotes storing large amounts of research data in the YODA vault, this is considerably cheaper than storing these data in the YODA research environment or the O-drive. YODA has additional functionality to improve the FAIRness of your data. You can archive your data with a standardized metadata form using YODA. Because data cannot be easily modified or deleted from the YODA archive and metadata are added, this is the best way to make sure your data are safely stored once your project ends. Storing data in the vault is also advised to secure raw data and make sure they cannot be accidentally deleted or modified. In addition, data that are stored in the archive can be published using YODA, thereby facilitating reuse of data. You can explore published datasets using DataCite Commons. YODA enables you to collaborate with other researchers, both inside and outside your own institution. The software is developed at Utrecht University and is used by multiple organisations, both in the Netherlands and abroad. All information how to use YODA is available on the UU website. The different ways to access data in YODA can be found here.

> **i** Storing data in Yoda
>
> Advantages:
>
> - Allows collaboration with external partners
> - Use the vault to protect data from accidental deletion or modification
> - Cost-effective for storage of large amounts of data in the vault
>
> Disadvantages:
>
> - Direct access to large files can be slow because they have to be transferred over the network

8.2 Organizing data

8.2.1 Folder structure

Your research folder ideally represents the whole lineage of all the data in your research project, from collection to publication. That typically includes the following categories:

- Raw data
- Code used to process the data
- Processed data (can be reproduced from raw data and code)
- Research output (e.g. figures, tables)
- Documentation, this includes:
    - Contracts, informed consent templates, etc.
    - Codebooks for questionnaires
    - Lab books

- Descriptions of parameters/settings for any measurement device

The contents of your data folder can be split in to types of files, data and metadata. Metadata and documentation provide information about your data, such as the topic of your project or the circumstances under which your data were obtained. Without metadata and documentation, a lot of data are just numbers without any meaning. Therefore, this "data about data" is essential for reliable scientific output and to understand, find, reuse and manage (the context of) your data. You can read more about metadata on the website of UU RDM Support.

The most rudimentary folders structure would therefore look something like this:

```
project_name/
|-- README.md
|-- data/
    |-- raw/
    |-- processed/
|-- code/
|-- documentation/
|-- output/
```

However, there is no one way to do it so the advice is to think about this before you start your project together with the data management representative from your research group.

Always add a README file to the top layer of your folder that explains how the folder structure is built up and what can be found in each of the subfolders. Also explain the file naming convention (see below) in this README file. In most cases it suffices to have one README file for the whole folder but sometimes it can help to add README files to subfolders.

i Important considerations for organizing folders in Yoda

Yoda has some **technical limitations** when it comes to organizing folders that should be taken into account when designing a folder structure:

- Only use lowercase letters, digits or hyphens for project folder names.
- Filenames within folder are case-sensitive and may also contain e.g. spaces (but never single quotes).
- File paths can be maximally 1023 characters long, always including the prefix '/nluu6p/home/research-' but the user-usable limit is lower due to how YODA works: at least 900 characters available NB: limits can differ between operating systems when trying to access YODA via e.g. WebDAV.

Some **additional recommendations** for naming research folders:

- Preferably as short as possible.
- Includes project name.
- Project name alone is hard to make meaningful (especially a short name that should last over time); so project folder names could also include the last name and initials of the researcher.
- A set project folder name structure allows for more overview when mounting YODA as a drive.
- Because of the way access rights work in YODA, often the same research project folder name is repeated with a suffix (e.g. adding '-shared') to share subsets of the data with different people.
- It would be very useful for continuity and easy transfer between data managers if every community contains a README file for data managers, describing the structure (i.e. community and subcommunity levels) and research project folder naming guidelines used in that community.

8.2.2 File naming convention

A good file naming structure helps with understanding what files contain and which version of the file you are working with. It's important to not make file names too long, however you should try to make the names as informative as possible. You can find some ideas for naming your files here. Again, it's good to think about naming your files before you start data collection and to discuss your ideas with the data management representative of your research group/department (see Section 4.1).

8.2.3 Data formats

There is a lot of diversity in data formats. When possible, it is important to choose for future-proof data formats (i.e. open, standardized formats) to be able to continue reading, displaying and processing research data in the future (e.g. CSV instead of XLSX). Furthermore, open data formats allow other researchers with different operating systems and different software packages to access the data.

Open data formats are preferred over proprietary formats because proprietary formats can become unsupported if the company that created them, stops developing the software or goes out of business. This is especially problematic with instruments that only output data in a proprietary file format.

There are many lists of open data formats to use as reference, not all of them contain the same guidance, and there can be many disagreements between researchers in the Open Science community. If you're going to be publishing data in a repository (see Chapter 12), they may provide a list of formats that you should use to publish there. If not, here are resources with open data formats:

- DANS list of open formats
- 4TU list (PDF)

8.3 Data breaches

Did something (possibly) happen with personal data that was not supposed to happen? Did anyone have unauthorised access? Did anyone who should have access lose that access? Have any personal data been lost? If so, that is considered a personal data breach. If you suspect a data breach, please report it as soon as possible via databreach@uu.nl.

For more information see:

- https://intranet.uu.nl/en/knowledgebase/what-do-i-have-to-do-if-there-is-a-personal-data-breach
- https://www.uu.nl/en/organisation/practical-matters/privacy/report-data-breach

8.4 Links with additional information

For further information please see the following links:

- https://www.uu.nl/en/research/research-data-management/tools-services/tools-for-storing-and-managing-data
- https://intranet.uu.nl/en/information-security-where-should-i-store-my-data
- https://intranet.uu.nl/kennisbank/onderzoeksinformatie-dgk (Dutch)

9 Documenting data

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10 Analyzing data

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10.1 Git and GitHub for code and software

10.1.1 FAIR code

The FAIR principles for data management can also be applied to code and software. Lamprecht et al. (2020) provide an academic framework for this. The Netherlands eScience Center in collaboration with DANS provide five actionable points that you can implement in your research work. These points are geared towards larger software projects, while researchers at IRAS usually create smaller scale projects in the form of scripts for which including all of these points might be excessive. In this paragraph we highlight the three points that are important for making your code FAIR in smaller scale projects and are easy to implement, provided you keep them in mind from the start of your project.

10.1.2 1. Use a publicly accessible repository with version control

At IRAS, we use Git along with the GitHub platform. Incorporating Git and GitHub into your project workflow has (at least) four distinct benefits:

- Your code is version-controlled, which means that you can easily track changes over time, and even undo them if needed.
- Multiple individuals can see and collaborate on the code, without clashing with each other and without sending code via email.
- GitHub provides a remote backup of your code, so it isn't stored only on your local machine.
- Your completed code has a permanent place to live, so you can easily share it with others or deploy it as a web page or an application.

If you connect your GitHub account to the Utrecht University GitHub organization, you can use all of GitHub's functionality without restriction. Utrecht University provides an organizational GitHub account where you can organize your code repositories and collaborate with others at the university. It is also possible to create teams within the organization. Currently, IRAS has one active group called IRAS-OneHealthChemical. More teams will be created in the future for other units, as they are needed.

When you create a GitHub repository for your research project, it is recommended that you set UtrechtUniversity as owner of the repository to gain the benefits of the organization. In principle, the copyright of work that you publish is owned by Utrecht University, but in daily practice you can make your own choices as if you were the copyright holder. You should also always add a license to the published work.

10.1.3 2. Add a license

Before publishing or sharing your code, add a software license to your repository. A license clearly states what others are allowed to do with your code, such as using, modifying, or redistributing it. Without a license, others have no permission to reuse your code, even if it is publicly available on GitHub. At IRAS, we encourage researchers to include an open-source license whenever possible. The choice of license depends on how you want your code to be reused. Choosealicense.com is a helpful website to learn which license suits your project the most. For most research projects, we recommend choosing one of the following licenses:

- MIT license: Short and highly permissive. Friction-free and most suited for a small library or script.
- Apache license 2.0: Highly permissive, but longer and more explicit. More suitable for larger software projects.
- GNU General Public License (GPL): A "copyleft" license that says that anything that is created using your code, should also use the same open rights as your code. With this license, closed source software using your code is not allowed to be distributed.

GitHub makes it easy to add a license when creating a repository or afterwards through the repository interface. Including a license from the start of your project makes it clear to collaborators and future users how your code can be used and shared. Keep in mind that your license should be consistent with any contractual obligations for the research that you are working on.

10.1.4 3. FAIR code checklist

Make sure you check all the boxes of UU-RDM FAIR Code cheatsheet before you publish your work!

10.1.5 Other helpful links

On the Utrecht University-RDM GitHub page, you can you can find many resources for getting started in writing FAIR research code:

- Best practices for git, GitHub, and research software development. Tailored for researchers of Utrecht University

Using Git and GitHub is an essential skill if you do any sort of programming work in your (academic) career. To learn the basics, the Research Data Management Support group at Utrecht University hosts workshops several times a per year, or on request:

- Introduction to GitHub

Part IV

After research

The current chapter describes:

- Chapter 11: how best to archive your research data after your research project.
- Chapter 12: how best to publish your research data after your research project.

11 Archiving data

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12 Publishing data

At the end of the project or when a paper has been submitted, you can publish your data in YODA or DataverseNL. Publishing your data fits the goals of Open Science but there can be reasons to not share all your data. However, you can choose to only publish the metadata of your research project, that way you can tell the world which data you have but keep control over who you share your data with and under what conditions. You can find more about publishing your data on the UU RDM Support website.

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Part V

Data security and privacy

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13 Legal instruments and agreements, policies and laws

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14 Handling personal data

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15 Data security

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16 …

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References

Bruce, R., and B. Cordewener. 2018. "Open Science Is All Very Well but How Do You Make It FAIR in Practice?" *LSE Impact of Social Sciences Blog.* London School of Economics & Political Science. https://blogs.lse.ac.uk/impactofsocialsciences/2018/07/26/open-science-is-all-very-well-but-how-do-you-make-it-fair-in-practice/.

Appendix A

List of mentioned tools

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Appendix B

Qualtrics tips & tricks

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